UNIVARIATE AND BIVARIATE ANALYSIS

Section A: Univariate Analysis (10 Questions)

1. How many records and columns are present in the cleaned crop-yield dataset? Display the first five rows and inspect the data types.

2. What are the minimum, maximum, mean, median, and standard deviation of Yield_kg_ha for the selected crops?

3. How many observations are available for Maize (corn), Rice, and Wheat individually?

4. What is the distribution of Yield_kg_ha for each crop? Create a histogram for each crop.

5. Which crop has the highest median yield, and which has the lowest median yield?

6. Are there zero-yield or negative-yield observations? Count them by crop.

7. Which 10 records have the highest recorded yields in the selected-crop dataset? Display Area, Item, Year, and Yield_kg_ha.

8. How does yield vary across years overall? Plot the number of observations available in each year.

9. For each crop, what are the 25th, 50th, and 75th percentiles of yield?

10. Are yield distributions strongly skewed? Plot boxplots for the three crops and describe any visible skew or extreme values.

Section B: Bivariate Analysis (10 Questions)

11. How does the distribution of yield differ between Maize (corn), Rice, and Wheat?

12. How has the average annual yield changed over time for each crop?

13. Which areas have the highest average yield for each crop over the available period?

14. For each crop, which areas have the lowest average yield, considering only areas with at least 10 observations?

15. Is there a relationship between the previous year's yield and the current year's yield?

16. How strongly are current yield and the two-year and three-year lagged yields related for each crop?

17. Which crop shows the largest year-to-year change in yield, and in which areas are the largest changes observed?

18. Do areas with higher average yields also show greater variability in yield?

19. For each crop, which area-year observations have the largest absolute difference from the previous year's yield?

20. Does the relationship between previous-year yield and current yield look different for Maize, Rice, and Wheat?